

TORRINGTON, WY, USA

WMO: 725763

Lat: 42.061N

Lon: 104.158W

Elev: 1280

StdP: 86.86

Time Zone: -7.00 (NAM)

Period: 01-19

WBAN: 94053

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-21.0	-17.2	-23.9	0.5	-20.0	-20.8	0.7	-15.5	14.8	4.2	13.0	3.1	2.3	300	0.641

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.0	36.1	17.4	34.2	17.2	32.6	17.1	20.3	29.6	19.5	29.0	18.7	28.0	4.6	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.7	14.8	22.8	16.8	14.0	22.3	15.9	13.2	21.7	63.9	29.3	60.7	28.9	58.1	27.9	23.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.8	11.2	9.7	-27.3	38.9	3.4	1.3	-29.8	39.8	-31.8	40.5	-33.7	41.2	-36.1	42.1		
(5)			-27.3	22.3	3.2	0.8	-29.6	22.8	-31.5	23.3	-33.2	23.7	-35.6	24.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.7	-1.5	-1.5	4.2	8.3	13.1	19.8	23.8	22.1	17.1	9.3	3.1	-2.2		
	DBStd	10.45	6.34	6.77	5.95	4.90	4.84	3.81	2.89	3.05	4.33	5.12	5.78	6.77		
	HDD10.0	1650	358	322	191	86	26	0	0	0	3	73	213	379		
	HDD18.3	3563	616	554	437	301	170	27	2	7	73	282	457	637		
	CDD10.0	1541	1	1	12	36	123	294	428	374	216	50	6	1		
	CDD18.3	412	0	0	0	1	9	72	172	123	36	1	0	0		
	CDH23.3	5794	0	0	12	47	242	1066	2134	1546	664	82	1	0		
	CDH26.7	2774	0	0	1	6	76	487	1153	753	280	17	0	0		
(14) Wind	WSAvg	4.2	4.4	4.5	4.7	5.0	4.6	4.2	3.8	3.7	3.7	4.0	4.0	4.3		
(15) Precipitation	PrecAvg	340	7	9	19	40	62	57	46	28	31	24	12	8		
	PrecMax	525	22	33	71	88	156	134	111	68	120	83	34	25		
	PrecMin	178	0	0	3	4	7	3	0	1	4	1	0	0		
	PrecStd	74	5	8	15	21	40	35	28	20	26	19	10	7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.1	18.9	25.1	27.5	31.8	37.2	38.8	36.7	34.7	28.8	22.3	15.6		
		MCWB	6.2	7.6	10.2	11.2	15.0	17.1	18.2	16.8	16.7	12.9	9.3	6.1		
	2%	DB	12.9	15.0	21.6	24.4	28.8	34.1	36.9	34.8	32.5	25.8	18.9	12.8		
		MCWB	4.6	5.8	8.7	10.6	14.5	16.6	17.8	17.0	15.4	11.7	7.9	4.8		
	5%	DB	10.7	12.1	18.0	22.0	26.2	32.0	35.0	33.1	30.1	22.7	16.2	10.4		
		MCWB	3.5	4.2	7.4	9.9	13.8	16.5	17.7	17.1	14.9	10.8	6.9	3.6		
	10%	DB	8.0	8.9	15.1	18.7	23.0	29.4	32.9	31.2	27.4	19.4	13.0	8.0		
		MCWB	2.4	2.7	6.2	8.9	12.7	16.0	17.8	17.1	14.5	9.7	5.3	2.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.8	7.9	10.7	12.8	17.8	20.7	21.7	21.0	19.0	14.1	10.1	6.7		
		MCDB	14.6	17.8	22.8	23.6	27.5	29.1	31.2	29.8	29.9	24.7	20.5	14.7		
	2%	WB	5.1	5.9	9.2	11.3	16.1	19.1	20.6	19.8	17.5	12.8	8.4	5.2		
		MCDB	12.1	14.0	19.8	21.8	24.5	28.1	29.9	29.0	27.4	22.8	18.2	12.3		
	5%	WB	3.8	4.6	7.8	10.3	14.9	18.0	19.9	18.9	16.4	11.5	7.1	3.9		
		MCDB	9.8	11.5	17.2	20.1	22.9	27.3	29.7	28.0	25.9	20.7	15.5	10.0		
	10%	WB	2.6	2.8	6.5	9.2	13.6	17.1	19.0	18.1	15.5	10.2	5.7	2.5		
		MCDB	8.0	8.4	14.6	17.9	21.7	26.5	28.6	27.1	24.1	18.4	12.8	7.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	13.5	13.6	15.0	14.9	14.7	16.5	17.0	17.0	17.2	16.0	14.9	13.2		
		MCDBR	16.4	17.5	20.8	21.4	19.9	20.4	20.4	19.9	21.6	22.3	19.8	16.2		
	5% WB	MCWBR	9.7	9.8	10.3	9.6	7.9	6.8	5.7	5.9	7.5	10.3	10.5	9.7		
		MCWBR	15.5	17.0	19.3	19.2	17.3	18.1	16.9	16.8	18.2	19.8	19.1	15.3		
(40) Clear-Sky Solar Irradiance	taub		0.245	0.255	0.277	0.305	0.325	0.335	0.360	0.354	0.317	0.278	0.259	0.247		
		taud	2.588	2.554	2.521	2.452	2.463	2.457	2.422	2.424	2.511	2.590	2.558	2.559		
	Ebn at Noon		936	973	981	966	948	935	907	904	922	931	906	902		
		Edh at Noon	65	79	92	107	109	110	113	109	93	75	66	61		
(44) All-Sky Solar Radiation	RadAvg		2.21	3.08	4.32	5.29	6.10	7.24	7.13	6.23	5.04	3.45	2.42	1.84		
	RadStd		0.09	0.19	0.32	0.33	0.51	0.41	0.28	0.19	0.29	0.29	0.13	0.10		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (15 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page